

Worksheet: Material balances in liquid-liquid extraction

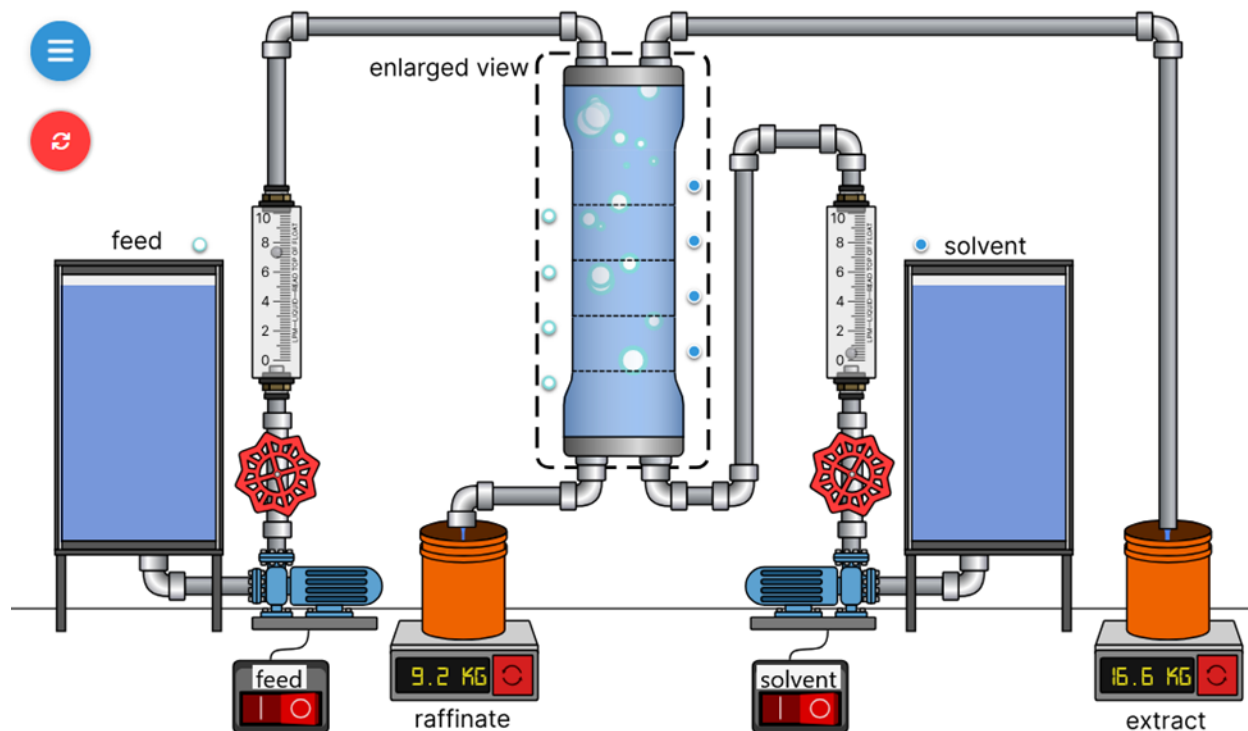
Name(s) _____

This experiment applies material balances to a liquid-liquid extraction process in which water as a solvent contacts an acetic acid/chloroform mixture in a steady-state countercurrent flow experiment. Because acetic acid is much more soluble in water, it is extracted from the chloroform organic phase into the aqueous phase.

Student learning objectives

1. Calculate total and component mass balances on a three-component system.
2. Use phase equilibrium data to partially generate the phase envelope on a ternary phase diagram.
3. Explain how raffinate and extract flow rates change from stage to stage.
4. Explain what happens during liquid-liquid extraction.

Equipment



The number of stages can be varied between three and five. In each stage, the upward-flowing aqueous solvent phase contacts the downward-flowing organic phase (chloroform, acetic acid, and a small concentration of water). The phases leaving each stage are in phase equilibrium. The water extracts acetic acid solute from the organic feed. Thus, the extract contains the water solvent with acetic acid and some chloroform. Chloroform is enriched in the remaining organic phase (raffinate) because acetic acid and water have been removed from it. Note that the feed and solvent tanks are large relative to the size of the column (thus the column is

enlarged in the simulation view). Thus, the tank volumes do not change significantly during an experiment. Two buckets that sit on scales collect the flows out of the column.

Notation

S = solvent mass flow rate

F = feed mass flow rate

E_i = mass flow rate of extract stream leaving stage i

R_i = mass flow rate of raffinate stream leaving stage i

x_{Fi} = mass fraction of component i in feed stream

x_{Si} = mass fraction of component i in solvent stream

x_{ni} = mass fraction of component i in raffinate stream leaving stage n

y_{ni} = mass fraction of component i in extract stream leaving stage n

ρ_i = density of component i .

ρ_{mix} = density of mixture

Questions to answer before starting the experiment

1. How many independent mass balances can you write for each stage of this three-component, liquid-liquid extraction? Explain.
2. The two phases contact each other in countercurrent flow. What might be the advantage of countercurrent flow over cocurrent flow?
3. Would you expect the feed stream (chloroform, acetic acid, water) to increase or decrease its flow rate as it flows down the extraction system? Why?
4. Would you expect the water solvent stream to increase or decrease its flow rate as it flows up through the extraction system? Why?

Before starting

Select the edit internals button just above the feed tank to choose the number of stages and record below.

Number of stages _____

Procedure:

1. Turn on the solvent pump by clicking the I/O switch to fill the column with solvent. Only after the column fills, turn on the feed pump. Adjust the feed and solvent valves to select flow rates where some acetic acid remains in the raffinate leaving the last stage. Record the two volumetric flow rates in Table 1. Use zoom with a scroll wheel to read the flow meter (L/min) more accurately.
2. The mass percentages of the three components in the feed and their densities are:

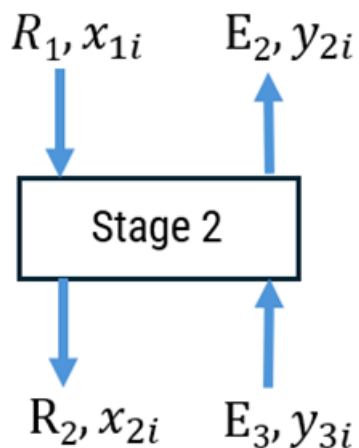
Component	Mass percent	Density (kg/L)
Acetic acid	52.0	1.49
Chloroform	43.0	1.04
Water	5.0	1.00

The solvent only contains water.

3. The density of a mixture is $\rho_{mix} = 1/\sum(x_i/\rho_i)$ where x_i is the mass fraction and ρ_i is the density of component i . Calculate fluid density in the feed tank and record it in Table 1.
4. Use the volumetric flow rates and densities to calculate the mass flow rates of the feed and the solvent and record the values in Table 1.
5. Allow the system to reach steady state. Then record the initial mass in extract bucket, collect flow for an interval of time, and record the time and mass change in Table 1. Repeat for the feed bucket. Calculate mass flow rates. Repeat the measurements and calculate averages in Table 1.

Table 1 Volumetric and mass flow rates

Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					F =
Solvent		1.00			S =
Extract (1 st)					E _{1,1st} =
Raffinate (1 st)					R _{n,1st} =
Extract (2 nd)					E _{1,2nd} =
Raffinate (2 nd)					R _{n,2nd} =
Extract (avg)					E _{1,avg} =
Raffinate (avg)					R _{n, avg} =



R_n and E_n are mass flow rates; x_{ni} and y_{ni} are mass fractions.

6. Measure the compositions of each stream by moving your mouse over the dots (white is raffinate, blue is extract) on the sides of the column. The top stage in the column is stage 1. As shown in the figure above, the aqueous phase leaving stage i is stream E_i and the organic phase leaving stage i is R_i .

Table 2 Mass fractions

	Mass fractions		
Stream	Acetic acid	Chloroform	Water
E ₁	$y_{1A} =$	$y_{1C} =$	$y_{1W} =$
E ₂	$y_{2A} =$	$y_{1C} =$	$y_{2W} =$
E ₃	$y_{3A} =$	$y_{3C} =$	$y_{3W} =$
E ₄	$y_{4A} =$	$y_{4C} =$	$y_{4W} =$
E ₅	$y_{5A} =$	$y_{5C} =$	$y_{5W} =$
S	0	0	$y_{SW} = 1.00$
F	$x_{FA} =$	$x_{FC} =$	$x_{FW} =$
R ₁	$x_{1A} =$	$x_{1C} =$	$x_{1W} =$
R ₂	$x_{2A} =$	$x_{2C} =$	$x_{2W} =$
R ₃	$x_{3A} =$	$x_{3C} =$	$x_{3W} =$
R ₄	$x_{4A} =$	$x_{4C} =$	$x_{4W} =$
R ₅	$x_{5A} =$	$x_{5C} =$	$x_{5W} =$

Data Analysis:

Do an overall mass balance on the system. Enter the mass flow rates of feed (F) and solvent (S) in Table 3 and add them together. Enter the mass flow rates of raffinate (R_n) and extract (E_1) out of the column in Table 3 and add them together. Calculate the difference and the percentage difference (relative to the total flow in) and enter those values in Table 3.

Table 3 Overall mass balances

Mass flow rates (kg/min)							
Mass flow in			Mass flow out				
F	S	F + S	R_n	E_1	$R_n + E_1$	Difference: in-out	% difference

Enter the flow rates in and out of the column in Table 4. Enter the mass fractions of the streams entering and leaving the column in Table 4 and then calculate the flow rates in and out of each species and the totals for each species and enter those values in Table 4. Calculate the differences (in – out) and enter those values in Table 4.

Table 4 Component mass flow rates

	F = kg/min	S = kg/min	mass flow rates in (kg/min)		
	x_{Fi}	y_{Si}	Fx_{Fi}	Sy_{Si}	$Fx_{Fi} + Sy_{Si}$
Acetic acid					
Chloroform					
Water					
	R_n = kg/min	E_1 = kg/min	mass flow rates out (kg/min)		
	x_{ni}	y_{1i}	$R_n x_{ni}$	$E_1 y_{1i}$	$R_n x_{ni} + E_1 y_{1i}$
Acetic acid					
Chloroform					
Water					
			Difference (in – out)		
			Acetic acid		
			Chloroform		
			Water		

Calculate the percent difference between in and out for each species (relative to the total flow in for each species).

Enter the flow rates F , S , R_n , and E_1 in Table 5. Write two component mass balances around stage 1 and solve these equations to calculate the R_1 and E_2 mass flow rates. Enter the values in Table 5. repeat this procedure for stage 2, then stage 3, etc. to determine all mass flow rates and enter them in Table 5.

Example mass balances for acetic acid for stage 1

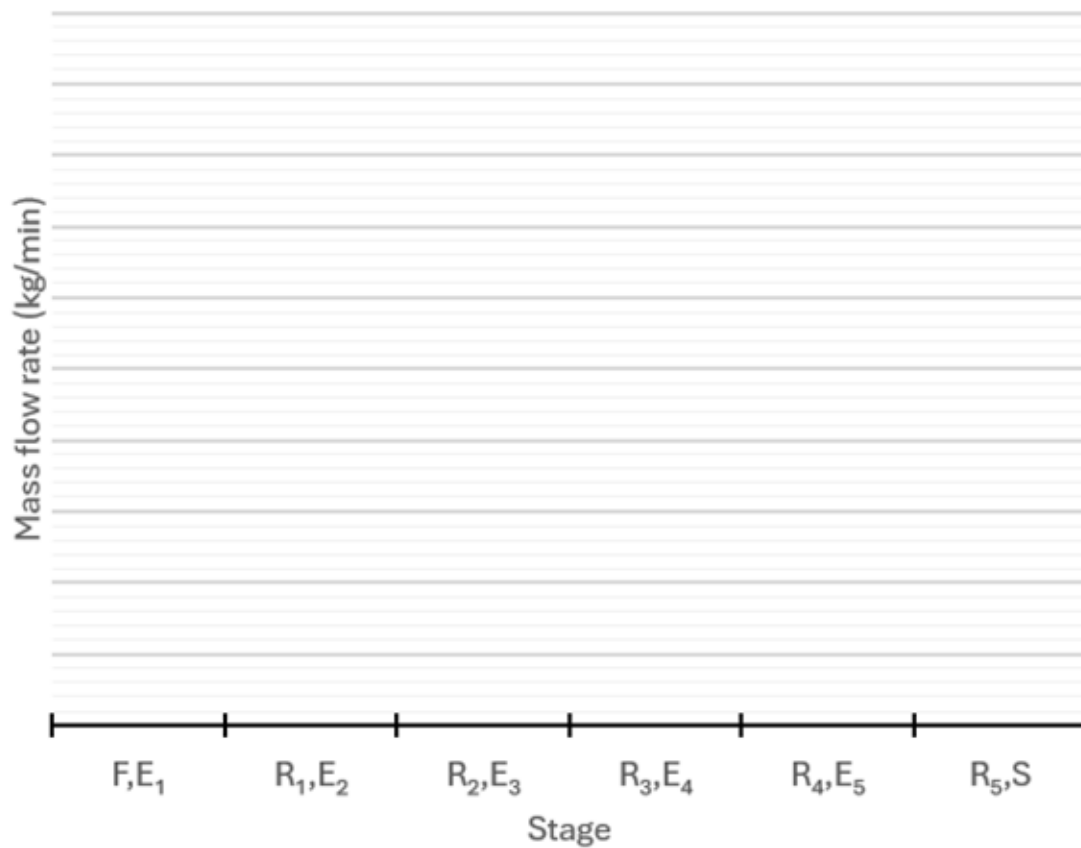
Flow in = flow out

$$F x_{FA} + E_2 y_{2A} = R_1 x_{1A} + E_1 y_{1A}$$

Table 5 Mass flow rates

Stream	Mass flow rate (g/s)
F	
R_1	
R_2	
R_3	
R_4	
R_5	
E_1	
E_2	
E_3	
E_4	
E_5	
S	

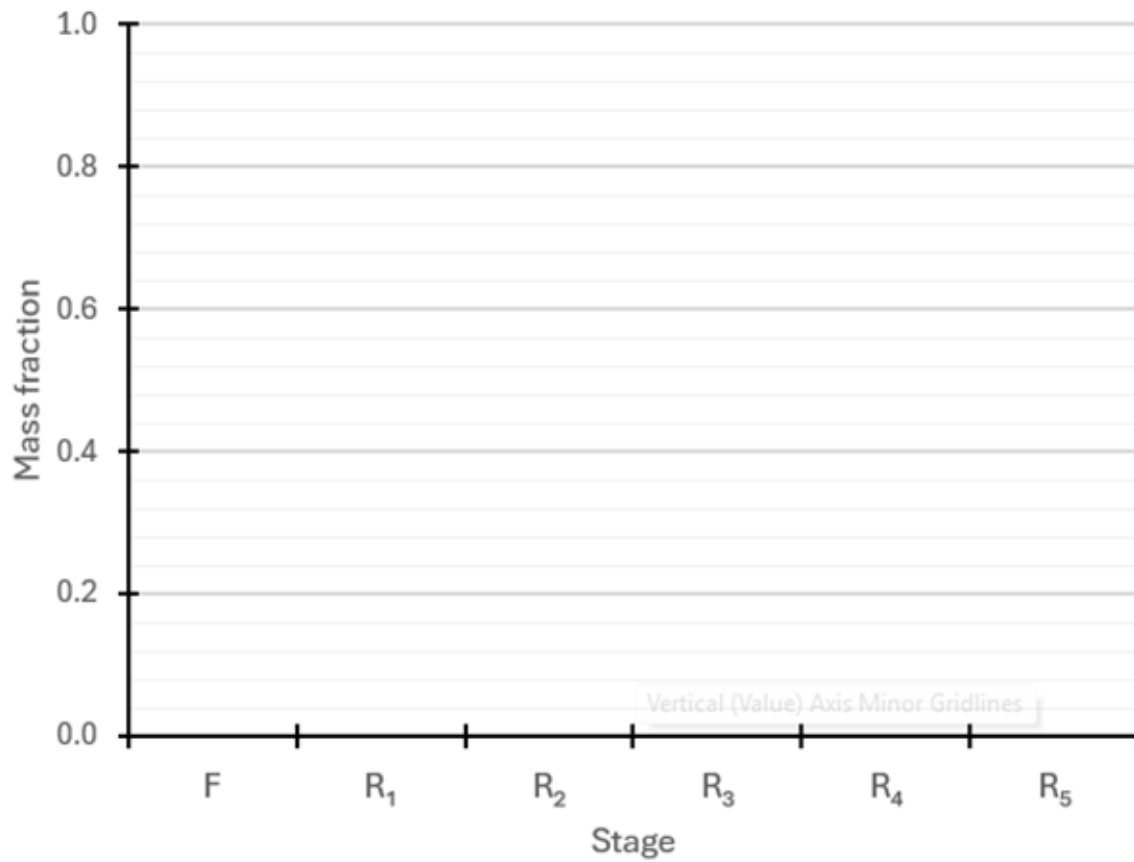
Plot mass flow rates of the streams on the y-axis versus the stage as indicated in the figure below.



Explain why the aqueous stream mass flow rates increase or decrease with stage number.

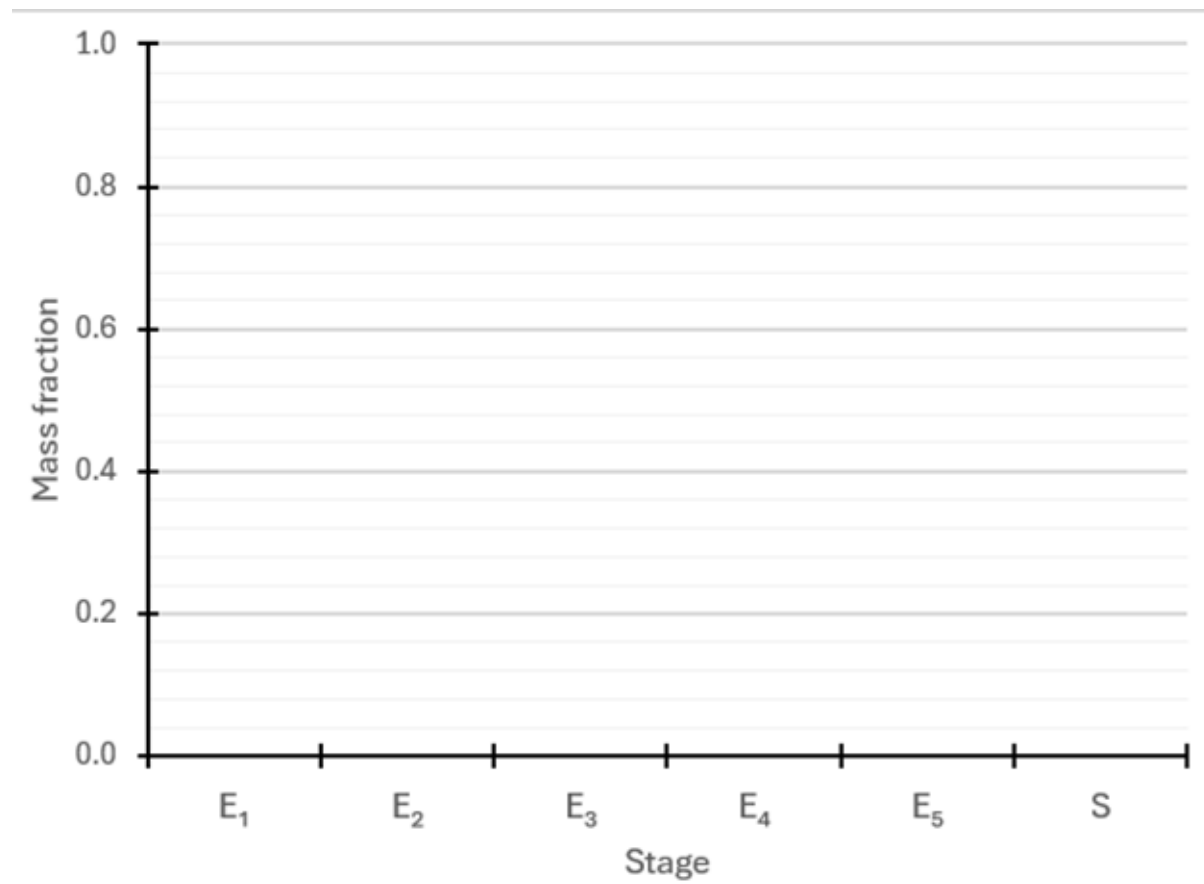
Explain why the organic stream mass flow rates increase or decrease with stage number.

Plot the mass fractions of each component for the feed flow stream.



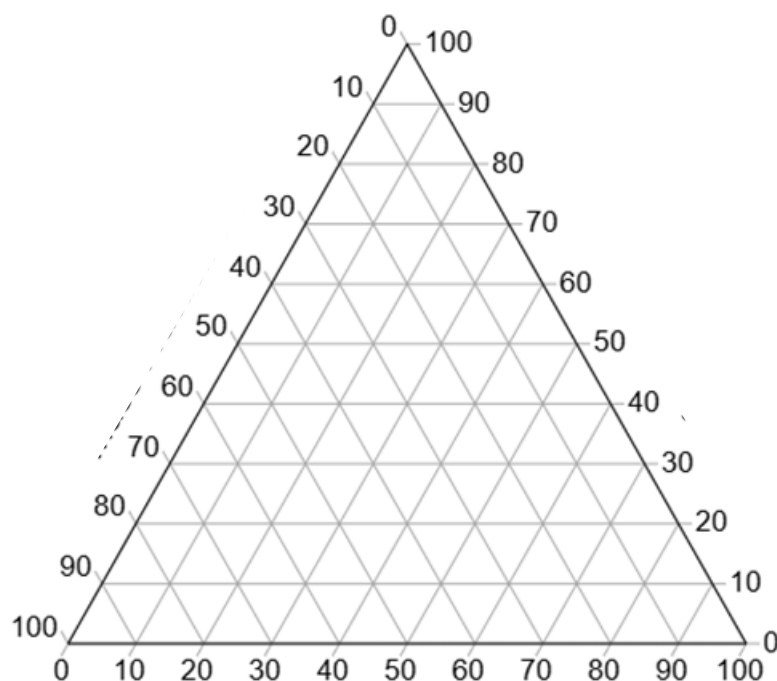
Which component transfers most strongly between phases.

Plot the mass fraction of each component for the solvent flow stream.



Explain the behavior in the graph.

The streams leaving each stage (e.g., E_2 and R_2 or stage 2) are in equilibrium. Tie lines connect two composition points that are in equilibrium on a ternary phase diagram. Points connected by tie lines are on the phase envelope of a ternary phase diagram. Locate the exit stream concentrations on the ternary phase diagram below and draw as much of the phase envelope as you can. Also draw the tie lines.



Questions to answer

1. Where might these measurements have errors?
2. What safety precautions would you take to conduct this experiment in the laboratory?